

ASAF BELILUS

Backend Software Engineer · Java & Spring Boot · Builder of SwimEdge

CONTACT

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[LinkedIn](#)
[GitHub](#)
[Portfolio](#)

SKILLS & TECHNOLOGIES

Languages

Java, Python, TypeScript, C/C++

Backend

Spring Boot, REST APIs, JWT Auth

Data

PostgreSQL, versioned schema migrations

Tools & Testing

Docker, Git, automated testing

Frontend

React, Vite

EDUCATION

M.Sc. Computer Science

Ben-Gurion University of the Negev
Researcher, Prof. Gera Weiss's Lab · 2026 – Present (Expected Oct 2027)
Current GPA 95.

B.Sc. Computer Science

Ben-Gurion University of the Negev · 2022 – 2025
Final GPA 84.2 · Active swimmer, BGU ASA competitive swim team.
Coursework: Operating Systems, Computer Architecture, Systems Programming, Databases

ACTIVITIES

Teaching Assistant — BGU · 2026 – Present

Supervises practicum coursework and grades/examines student papers.

Individual/Group Swimming Instructor — Wingate Institute · 2014 – Present

Teaches fundamentals and advanced technique to swimmers of all ages.

Volunteer Math & CS Tutor — BGU · 2026 – Present

Tutors first-year students in math and CS, focused on reservist students returning to their studies.

LANGUAGES

Hebrew — Native
English — Fluent

PROFILE

Computer Science M.Sc. student seeking a backend software engineering student position. Built SwimEdge's Spring Boot backend as 40 domain services behind 31 REST controllers, with six-role authorization, 23 Flyway migrations, and tests against PostgreSQL. Brings the same debugging discipline to cross-faculty computer-vision research.

RESEARCH

Swimmer Biomechanics Research — Prof. Gera Weiss's Lab

Ben-Gurion University of the Negev · in collaboration with Prof. Raziell Reimer, Faculty of Engineering · 2026 – Present

- Conducting cross-disciplinary research applying computer vision and control theory to competitive-swimming biomechanics, bridging the Computer Science and Engineering faculties.
- Built Python pipelines that reconstruct swimmer joint motion from 3-D underwater pose-estimation data, using forward/inverse kinematics and nonlinear least-squares optimization to drive a third-party hydrodynamic simulation engine.
- Diagnosed systematic model-fitting errors (degenerate optimization bounds, skeletal geometry mismatches) via quantitative probing against ground-truth motion-capture data, backed by automated regression testing.

FEATURED PROJECT

SwimEdge — Full-Stack Competitive Swimming Platform

Founder & Sole Engineer · 2025 – Present · [GitHub](#)

- Designed and built a full-stack platform for competition management and performance analytics, motivated by a gap identified firsthand as a competitive swimmer — owning backend and infrastructure end to end.
- Engineered a modular, service-oriented Spring Boot + PostgreSQL backend, profiling and optimizing critical paths for runtime speed and memory footprint, and evolving the schema through 23 versioned migrations.
- Built a custom parser converting official Israeli Swimming Association PDF/Excel results into clean relational data. Opened direct discussions with the Israel Swimming Association and engaged Israel's national sports administration (Ministry of Culture and Sport) on the opportunity for technological modernization across sport federations.

MILITARY SERVICE

Combat Commander, Paratrooper Brigade, IDF

2016 – 2019

- Trained and mentored newly recruited soldiers; led operations and managed company-level logistics.

Reservist, Paratrooper Brigade, IDF

2019 – Present

- Lead frontline teams under pressure, maintaining calm and accountability.